



Adafruit-ILI9341:

```
$> pip install Adafruit-GPIO
$> pip install Adafruit-ILI9341
```

4. Install Pillow (for Image, ImageDraw, ImageFont):

```
$> pip install pillow
```

5. For USB microphone use:

```
$> pip install pyaudio
```

6. Install googletrans (Translator):

```
$> pip install googletrans==4.0.0-rc1
```

7. Install RPi.GPIO (in all likelihood, this is already installed in your Pi zero, if not):

```
$> sudo apt-get install python3-rpi.gpio
```

Next, the sound input and output need to be configured.

Sound output configuration on Pi Zero

Since Pi Zero lacks built-in sound output, the following steps are essential for the sound output configuration:

1. Use any two of GPIO 12, 13, 18 or 19 pins to force audio output.
2. Add a line to the `/boot/config.txt` file.
3. Open the file, add the line at the bottom, save, and reboot.
4. After rebooting, audio will output through the selected GPIO pins.
5. Connect headphones to the IN-L and IN-R pins for sound output..
6. Use PAM8403 amplifier to enhance the sound for the speaker:

```
dtoverlay=audremap,pins_18_13
```

Overclocking of Pi Zero

In the same `~/boot/config.txt` file, locate the overclocking section, comment out the existing configuration, and add the lines below. Save and reboot the Pi. It will then operate at 1.3GHz. Attach a good metallic heatsink to the CPU and monitor its temperature using the following command: `[vcgencmd measure_temp]`

```
#uncomment to overclock the arm. 700 MHz
is the default.
```

```
#arm_freq=800
#Bera wrote these two lines
arm_freq=1300
over_voltage=2
```

Voice input to Pi Zero

Voice output is functioning perfectly. To enable voice input, a cost-effective USB microphone from robu.in, priced around 170 rupees (\$2), is selected. The microphone works seamlessly out of the box. Insert it into the only USB port of the Pi Zero, and run the following command to identify the hardware device number:

```
$> lsusb output ...
Bus 001 Device 002: ID 4c4a:4155 Jieli
Technology UACDemoV1.0
Bus 001 Device 001: ID 1d6b:0002 Linux
Foundation 2.0 root hub
$> arecord -f S16_LE -r 33100 -d 10 -c 1
output.wav
```

- `-f S16_LE`: Specifies the audio format. S16_LE stands for signed 16-bit little-endian.
- `-r 33100`: Sets the sample rate to 33100Hz.
- `-d 10`: Specifies the duration of the recording in seconds (e.g., 10 seconds in this example).
- `-c 1`: Sets the number of audio channels to 1 (mono). Adjust as needed for stereo recordings (`-c 2` for stereo).
- `output.wav`: Specifies the output file name (output.wav in this example).

After running this command, the scorecard will record audio at a sample rate of 33100Hz for 10 seconds and save it to the output.wav file in the current directory. Adjust the options (`-d` for duration, output file name, etc) according to your requirements.

To play the audio file `$> aplay output.wav`, will play your invaluable 10 seconds acoustic creation. Adjust the mike suitably to get clear audio recordings. Keep the speakers away from the mike to avoid feedback problems.

Internet connection

Ensure the Pi Zero is connected to Wi-Fi or another internet source. Use the following command to access the configuration tool:

```
$> sudo raspy-config and ;
```

Navigate to 'Network Options,' enter the SSID and password, exit the tool, and reboot. Upon reboot, the Pi Zero will automatically connect to the Wi-Fi network. Alternatively, manually edit the configuration file:

```
$> sudo nano /etc/wpa_supplicant/wpa_supplicant.conf ; in the
file do the following-
```

```
country = IN # (replace 'IN' with
your country code IN = India)
```

```
network={
    ssid="your_network_name"
    psk="your_password"
}
```

After completing these configurations, download the provided code from (<http://electronicsforyou.com>).

Circuit

Fig. 2 illustrates the circuit diagram for the Pi Zero portable translator. The system includes a Raspberry Pi Zero (MOD 1), an SPI TFT display (MOD 2), a PAM8403 module (MOD 3), a USB microphone, and other essential components. Connect these as depicted in Fig. 2 and Fig. 3.

Two push-to-on switches are connected to GPIO pins 17 and 22 for 'English to Hindi' (en-hi) and 'Hindi to English' (hi-en) conversions, respectively. The 7.1cm LCD panel (320x240 pixels) is connected via SPI to display operation sequences. For smaller TFT displays (e.g., 3.65cm, 128x128 pixels), adjust the pixel size in the code for compatibility.

Project command

Two push buttons handle all operations. Press the 'en-hi' button to prepare the device for English voice input. After speaking a short English